

Equivalência de Domínios — UStAI ↔ Synthetic User Stories (ReFAIR)

Documento gerado para o projeto **IN-0953-ReFair**. Estabelece a correspondência entre o eixo de "domínio" do dataset **UStAI-annotated_V2** e os eixos de **domínio de aplicação** e **ML task** do dataset **Synthetic User Stories** do ReFAIR.

1. Objetivo

Mapear cada *user story* do dataset **UStAI** para o vocabulário do dataset **Synthetic User Stories** (ReFAIR), produzindo um dataset de equivalência que permita usar as user stories do UStAI como entrada/validação externa da pipeline do ReFAIR (ver `documents/analise-replicacao-novo-dataset.md`).

2. Os dois datasets e a coluna de "domínio"

Dataset	Arquivo	Coluna usada como "domínio"	Cardinalidade
UStAI	UStAI-annotated_V2.pdf	role_shorten (papéis declarados na user story)	1260 user stories · 179 papéis distintos · 42 <i>abstracts</i>
Synthetic User Stories	Synthetic User Stories.xlsx	Domain (domínio de aplicação)	34 domínios
Synthetic User Stories	Synthetic User Stories.xlsx	Machine Learning Task	367 ML tasks

⚠ **Ressalva importante.** O PDF `Synthetic-User-Stories.pdf` entregue contém, na verdade, a lista das **367 ML Tasks** (coluna `Machine Learning Task` / aba *ML Dictionary*) — **não** a coluna de domínio. A coluna `Domain` real do dataset Synthetic está na aba *Dataset* do `.xlsx` e tem **34 domínios de aplicação**. Por isso a equivalência foi feita para **os dois eixos** (domínio de aplicação e ML task), conforme combinado.

Os **34 domínios** do Synthetic (ontologia de Fabris et al., reusada pelo ReFAIR):

Biology, Cardiology, Computer Networks, Computer Vision, Demography, Dermatology, Economics, Education, Endocrinology, Finance & Marketing, Health, Information Systems, Law, Library, Linguistics, Literature, Medicine, Movies, Music, Nephrology, News, Pediatrics, Pharmacology, Plant Science, Political Science, Psychology, Radiology, Social Media, Social Networks, Social Work, Sociology, Sport, Transportation, Urban Studies

A classificação de **ML task** no ReFAIR é **multi-label**: cada user story recebe um *conjunto* de tasks. As 367 ML tasks *fine-grained* são, por isso, agregadas em **25 labels de alto nível** pela aba *task augmentation mapping* do mesmo `.xlsx` — esses 25 labels são o alvo real do classificador de tasks (ver Seções 3 e 4):

Classification, Regression, Ranking, Matching, Risk Assessment, Representation Learning, Clustering, Anomaly Detection, Districting, Task Assignment, Spatio-Temporal Process Learning, Graph Diffusion, Graph Augmentation, Resource Allocation, Subset Selection, Data Summarization, Graph Mining, Pricing, Advertising, Entity Resolution, Sentiment Analysis, Bias Detection in Word Embeddings, Bias Detection in Language Models, Machine Translation, Speech Recognition

3. Metodologia da equivalência

- Extração do UStAI.** As 1260 user stories foram extraídas do PDF (42 *abstracts* × 3 LLMs × 10 US). Cada linha tem `id`, `abstract`, `us_num`, `llm`, `role_shorten` e o texto da *user_story*.
- **Âncora = role_shorten.**** O `role_shorten` é o campo-chave do lado UStAI, como solicitado.
- *Desambiguação pelo abstract.** Cada abstract (A1–A42) descreve **um único sistema de IA** — logo, todas as 30 user stories de um abstract* compartilham o mesmo domínio de aplicação e a mesma

ML task. Papéis genéricos (`researcher`, `developer`, `data scientist`, `manager`, `user`...) aparecem em dezenas de *abstracts*; sozinhos não determinam um domínio. Por isso o domínio é fixado **por abstract** (o contexto do sistema), exatamente a desambiguação escolhida. Papéis específicos de domínio (`radiologist`, `cardiologist`, `taxi driver`...) são coerentes com o *abstract* em que ocorrem.

4. **Domínio de aplicação.** Cada *abstract* foi classificado em 1 dos 34 domínios por análise temática das suas user stories.
5. **ML task — classificação multi-label.** No ReFAIR a task de ML **não é rótulo único**: o classificador de tasks é **multi-label** (Linear SVC + Label Powerset — ver `documents/analise-replicacao-novo-dataset.md`, Seção 1). Por isso a equivalência de ML task tem **duas camadas**:
 - **ML task âncora (fine-grained).** Cada *abstract* foi associado a 1 das 367 tasks do vocabulário controlado: a técnica nomeada quando o estudo gira em torno dela (ex.: *Fuzzy SVM* → `support vector machine`, *GANs* → `generative adversarial network`), ou o conceito da tarefa quando não há técnica única (ex.: *anomaly detection*, *recommendation system*). É a **ponte** para o vocabulário do Synthetic.
 - **Labels multi-label.** A task âncora é então **expandida** para um **conjunto de labels de alto nível** usando a aba ****task augmentation mapping**** do próprio *Synthetic User Stories.xlsx* — exatamente o mecanismo com que o ReFAIR constrói o *ground truth* multi-label do classificador de tasks. São **25 labels** possíveis; cada *abstract* recebe de 1 a 25 labels (ver Seção 4).
6. **Confiança.** Cada *abstract* recebe *High/Medium/Low* para o mapeamento de domínio. *Low* = não há domínio perfeito entre os 34 (ex.: A4 militar, A38 recrutamento).

4. Tabela de equivalência por *abstract* (núcleo da equivalência)

Cada *abstract* = 30 user stories (10 Gemini_1.5_flash + 10 Llama 3.1 70b + 10 O1_mini). A coluna **ML Task âncora** é a task *fine-grained* (1 das 367) que melhor nomeia a técnica do estudo de origem; ela é expandida para o conjunto **multi-label** de labels de alto nível logo abaixo.

Abstract	Tema do sistema de IA	Domínio equivalente (Synthetic)	ML Task âncora (fine-grained)	Confiança
A1	Autonomous vehicles (AI + real-time data for perception & collision avoidance)	Transportation	machine perception	High
A2	Wearable-based detection of physical aggression in children	Psychology	time-series classification	Medium
A3	Real-time aggression detection on Twitter	Social Media	text classification	High
A4	AI-powered military drones & international security	Political Science	reinforcement learning	Low
A5	GoCart autonomous delivery robot in healthcare/residential facilities	Health	robot learning	High
A6	ADAS driver-intention recognition with Gaussian Mixture Models	Transportation	mixture model	High
A7	ML-based mental health / mental illness prediction	Psychology	classification method	Medium
A8	Anomaly detection in smart-health wearable data	Health	anomaly detection	High
A9	Heart disease diagnosis with Fuzzy SVM & incremental learning	Cardiology	support vector machine	High
A10	Blood-glucose / hypoglycemia prediction with Support Vector Regression	Endocrinology	support vector regression	High

A11	Anti-counterfeiting copy-detection patterns (digital twin, image-to-image GANs)	Computer Vision	generative adversarial network	Medium
A12	Firearm Violence Vulnerability Index (XGBoost community-risk forecasting)	Social Work	gradient boosting	Low
A13	AI-DrugNet drug repurposing for Alzheimer's disease	Pharmacology	deep learning	High
A14	NYC taxi / Uber ridership prediction	Transportation	forecasting algorithms	High
A15	Income / poverty prediction from demographic & lifestyle data	Economics	classification method	High
A16	Newborn cry diagnostic system (cry-audio classification)	Pediatrics	classification method	High
A17	Sleep apnea / hypopnea prediction with LAMSTAR neural networks	Medicine	artificial neural network	High
A18	Tuberculosis detection from chest X-rays with CNNs	Radiology	convolutional neural network	High
A19	Credit-card default prediction	Finance & Marketing	classification method	High
A20	Drug-abuse prediction from Big Five personality traits	Psychology	classification method	Medium
A21	Racial & gender bias in online freelancing marketplaces	Economics	data mining	Low
A22	Multidimensional freelancer assessment framework for online labor	Economics	learning ranking	Low
A23	Search-engine bias analysis (PAWS platform)	Information Systems	information retrieval	Medium
A24	ArnetMiner academic search / academic social network	Library	information retrieval	High
A25	E-commerce recommendation (item-to-item collaborative filtering)	Information Systems	recommendation system	Medium
A26	Explainable recommendation justifications (Seq2Seq, aspect-planning)	Information Systems	natural language generation	Medium
A27	Session-based recommendation with RNNs	Information Systems	recommendation system	Medium
A28	Reciprocal dating recommender from speed-dating data	Social Networks	recommendation system	Medium
A29	Historical census record linkage	Demography	classification method	High
A30	Recidivism prediction with Decision Trees (criminal justice)	Law	decision tree	High
A31	Advertisement identification on web pages	Finance & Marketing	classification method	Low
A32	Stance detection with multi-task learning (NLP)	Linguistics	multi-task learning	Medium
A33	Semantic news search with Wikidata annotation	News	information retrieval	High
A34	DNN-based network intrusion detection system	Computer Networks	deep neural network	High
A35	Wholesale sales prediction with ARIMA	Finance & Marketing	time series forecasting	High

A1	machine perception	Classification; Regression; Ranking; Representation Learning; Clustering; Anomaly Detection; Spatio-Temporal Process Learning; Graph Mining; Sentiment Analysis; Bias Detection in Word Embeddings; Bias Detection in Language Models; Machine Translation; Speech Recognition	13
A2	time-series classification	Classification; Representation Learning; Clustering; Anomaly Detection; Subset Selection	5
A3	text classification	Sentiment Analysis; Bias Detection in Language Models; Machine Translation; Speech Recognition	4
A4	reinforcement learning	Districting; Task Assignment; Resource Allocation; Pricing; Advertising	5
A5	robot learning	Districting; Task Assignment; Resource Allocation	3
A6	mixture model	Classification; Clustering; Anomaly Detection; Graph Mining	4
A7	classification method	Classification	1
A8	anomaly detection	Anomaly Detection	1
A9	support vector machine	Classification; Regression; Ranking; Anomaly Detection; Entity Resolution; Sentiment Analysis; Bias Detection in Word Embeddings; Bias Detection in Language Models; Machine Translation; Speech Recognition	10
A10	support vector regression	Regression	1
A11	generative adversarial network	Representation Learning; Anomaly Detection; Data Summarization; Entity Resolution; Bias Detection in Language Models	5
A12	gradient boosting	Classification; Regression; Ranking; Representation Learning; Clustering; Anomaly Detection; Districting; Graph Mining; Pricing; Advertising; Entity Resolution; Sentiment Analysis; Bias Detection in Word Embeddings; Bias Detection in Language Models	14
A13	deep learning	Classification; Regression; Ranking; Matching; Risk Assessment; Representation Learning; Clustering; Anomaly Detection; Districting; Task Assignment; Spatio-Temporal Process Learning; Graph Diffusion; Graph Augmentation; Resource Allocation; Subset Selection; Data Summarization; Graph Mining; Pricing; Advertising; Entity Resolution; Sentiment Analysis; Bias Detection in Word Embeddings; Bias Detection in Language Models; Machine Translation; Speech Recognition	25
A14	forecasting algorithms	Regression	1
A15	classification method	Classification	1
A16	classification method	Classification	1
A17	artificial neural network	Classification; Regression; Representation Learning; Clustering; Anomaly Detection; Spatio-Temporal Process Learning; Graph Diffusion; Graph Augmentation; Subset Selection; Data Summarization; Graph Mining; Sentiment Analysis; Bias Detection in Word Embeddings; Bias Detection in Language Models; Machine Translation; Speech Recognition	16
A18	convolutional neural network	Classification; Representation Learning; Clustering; Anomaly Detection; Data Summarization; Graph Mining; Entity Resolution	7
A19	classification method	Classification	1

A20	classification method	Classification	1
A21	data mining	Classification; Regression; Clustering; Anomaly Detection; Data Summarization; Graph Mining; Entity Resolution	7
A22	learning ranking	Ranking; Advertising	2
A23	information retrieval	Ranking; Matching; Pricing; Advertising; Entity Resolution; Sentiment Analysis; Bias Detection in Language Models	7
A24	information retrieval	Ranking; Matching; Pricing; Advertising; Entity Resolution; Sentiment Analysis; Bias Detection in Language Models	7
A25	recommendation system	Ranking	1
A26	natural language generation	Classification; Regression; Representation Learning; Clustering; Sentiment Analysis	5
A27	recommendation system	Ranking	1
A28	recommendation system	Ranking	1
A29	classification method	Classification	1
A30	decision tree	Classification; Regression; Anomaly Detection; Subset Selection; Data Summarization	5
A31	classification method	Classification	1
A32	multi-task learning	Classification; Regression; Ranking; Matching; Risk Assessment; Representation Learning; Anomaly Detection; Task Assignment; Spatio-Temporal Process Learning; Graph Diffusion; Graph Augmentation; Entity Resolution; Sentiment Analysis; Bias Detection in Word Embeddings; Bias Detection in Language Models; Machine Translation; Speech Recognition	17
A33	information retrieval	Ranking; Matching; Pricing; Advertising; Entity Resolution; Sentiment Analysis; Bias Detection in Language Models	7
A34	deep neural network	Classification; Regression; Ranking; Representation Learning; Anomaly Detection; Spatio-Temporal Process Learning; Graph Diffusion; Graph Augmentation; Subset Selection; Data Summarization; Graph Mining; Entity Resolution; Sentiment Analysis; Bias Detection in Word Embeddings; Bias Detection in Language Models; Machine Translation; Speech Recognition	17
A35	time series forecasting	Regression; Anomaly Detection; Spatio-Temporal Process Learning; Graph Diffusion; Resource Allocation; Subset Selection	6
A36	time-series classification	Classification; Representation Learning; Clustering; Anomaly Detection; Subset Selection	5
A37	data mining	Classification; Regression; Clustering; Anomaly Detection; Data Summarization; Graph Mining; Entity Resolution	7
A38	classification method	Classification	1
A39	classification method	Classification	1
A40	anomaly detection	Anomaly Detection	1

A41	attention mechanism	Representation Learning; Sentiment Analysis; Machine Translation; Speech Recognition	4
A42	recommendation system	Ranking	1

Leitura. A task âncora é o termo *fine-grained* do estudo de origem; os **labels multi-label** são o que o classificador do ReFAIR realmente prevê. Técnicas de uso geral expandem para conjuntos amplos — *deep learning* (A13) cobre os 25 labels, *multi-task learning* (A32) e *deep neural network* (A34) cobrem 17, *artificial neural network* (A17) cobre 16. Técnicas específicas expandem para conjuntos enxutos — *support vector regression* (A10) → só Regression; *anomaly detection* (A8, A40) → só Anomaly Detection. Ver ressalva na Seção 8.

Distribuição das ML task labels (multi-label) nas 1260 user stories

Como a classificação é multi-label, cada user story recebe **em média 5,3 labels** (1 a 25). A soma das frequências abaixo é **6720** — bem maior que 1260, justamente por ser multi-label.

Label de ML task (multi-label)	Nº user stories	Nº abstracts
Classification	720	24
Anomaly Detection	540	18
Regression	420	14
Ranking	420	14
Sentiment Analysis	390	13
Representation Learning	360	12
Entity Resolution	360	12
Bias Detection in Language Models	360	12
Clustering	330	11
Graph Mining	270	9
Data Summarization	240	8
Machine Translation	240	8
Speech Recognition	240	8
Subset Selection	210	7
Advertising	210	7
Bias Detection in Word Embeddings	210	7
Spatio-Temporal Process Learning	180	6
Pricing	180	6
Matching	150	5
Graph Diffusion	150	5
Districing	120	4
Task Assignment	120	4
Graph Augmentation	120	4

Resource Allocation	120	4
Risk Assessment	60	2

Distribuição das 1260 user stories por domínio equivalente

Domínio	Nº user stories	Nº abstracts
Information Systems	120	4
Psychology	120	4
Transportation	120	4
Economics	90	3
Finance & Marketing	90	3
Health	90	3
Computer Vision	60	2
Social Networks	60	2
Cardiology	30	1
Computer Networks	30	1
Demography	30	1
Endocrinology	30	1
Law	30	1
Library	30	1
Linguistics	30	1
Medicine	30	1
Music	30	1
News	30	1
Pediatrics	30	1
Pharmacology	30	1
Political Science	30	1
Radiology	30	1
Social Media	30	1
Social Work	30	1
Sport	30	1

5. Equivalência `role_shorten` → domínio

Dos **179** papéis distintos, **110** mapeiam para um **único** domínio (papéis específicos) e **69** aparecem em **vários** domínios (papéis genéricos) — confirmando que o papel isolado não determina o domínio e que a desambiguação pelo *abstract* é necessária.

5.1. Papéis com domínio único

Healthcare Administrator	1	Health
healthcare policy advisor	1	Medicine
historian	2	Demography
human-factors specialist	1	Transportation
innovator	1	Computer Vision
interoperability coordinator	1	Political Science
job seeker	2	Psychology
judge or magistrate	1	Law
kitchen staff member	1	Health
knowledge base specialist	1	News
lab technician	3	Health
law enforcement agency	1	Law
linguist	2	Linguistics
machine learning model	1	Psychology
member of the Pokec social network community	1	Social Networks
military commander	3	Political Science
modeler	1	Library
moderator	3	Social Media
music enthusiast	7	Music
music industry professional	2	Music
music label	1	Music
music producer	1	Music
music streamer	1	Music
musician	1	Music
neonatologist	3	Pediatrics
Network engineer	1	Transportation
network security specialist	1	Transportation
neurologist	3	Pharmacology
news agency	3	News
News editor	1	News
non-profit organization leader	1	Economics
non-proliferation expert	1	Political Science
OEM	1	Transportation
online retailer	7	Information Systems

organizer	2	Social Networks
participant	1	Transportation
passenger	3	Transportation
peacekeeper	1	Political Science
pediatrician	1	Pediatrics
person with dietary restrictions	1	Computer Vision
pharmaceutical company scientist	1	Pharmacology
pharmacist	1	Pharmacology
player	9	Sport
playlist creator	1	Music
playlist curator	1	Music
printing press operator	1	Computer Vision
public transportation official	1	Transportation
publisher	1	Library
radiologist	5	Radiology
recruiter	3	Psychology
research evaluator	1	Library
research funder	1	Library
resident	1	Health
residential care staff member	2	Health
restaurant operator	1	Computer Vision
restaurant owner	1	Computer Vision
safety expert	3	Transportation
school counselor	1	Psychology
security expert	3	Computer Vision
shareholder of a wholesale company	1	Finance & Marketing
single person	2	Social Networks
sleep specialist	2	Medicine
social media moderator	1	Linguistics
social scientist studying the impact of technology on dating	1	Social Networks
sociologist studying assimilation and discrimination	1	Demography
software architect	1	Information Systems
sports fan	8	Sport
system	1	Health

taxi dispatcher	2	Transportation
taxi driver	1	Transportation
tester	1	Transportation
therapist	1	Psychology
traffic engineer	1	Transportation
trainer	1	Political Science
vaccination site coordinator	1	Health
visually impaired person	1	Transportation

5.2. Papéis multi-domínio (genéricos)

Formato: Domínio (nº user stories).

role_shorten	Nº US	Domínios equivalentes
researcher	162	Information Systems (15), Psychology (14), Economics (14), Transportation (13), Computer Vision (11), Health (9), Social Work (7), Pharmacology (6), Medicine (6), Library (6), Social Networks (6), Linguistics (6), Computer Networks (6), Cardiology (5), Pediatrics (5), Finance & Marketing (5), Demography (5), Social Media (4), Endocrinology (4), Radiology (4), News (4), Law (3), Political Science (2), Sport (1), Music (1)
developer	139	Information Systems (30), Computer Vision (11), Finance & Marketing (11), Transportation (10), Psychology (10), Health (9), Social Networks (6), Pediatrics (5), Medicine (5), Social Media (4), Cardiology (4), Endocrinology (4), Library (4), News (4), Computer Networks (4), Law (3), Linguistics (3), Radiology (2), Economics (2), Demography (2), Sport (2), Political Science (1), Social Work (1), Pharmacology (1), Music (1)
data scientist	81	Information Systems (9), Psychology (7), Transportation (7), Economics (7), Health (6), Finance & Marketing (5), Social Networks (4), Cardiology (3), Computer Vision (3), Pharmacology (3), Library (3), Demography (3), Law (3), Computer Networks (3), Social Media (2), Social Work (2), Pediatrics (2), Medicine (2), Linguistics (2), Music (2), Endocrinology (1), Radiology (1), News (1)
analyst	56	Finance & Marketing (9), Transportation (6), Social Networks (6), Psychology (4), Economics (4), Information Systems (4), Linguistics (4), Health (2), Computer Vision (2), Social Work (2), Demography (2), Computer Networks (2), Sport (2), Music (2), Political Science (1), Cardiology (1), Endocrinology (1), Law (1), News (1)
manager	45	Finance & Marketing (13), Transportation (8), Health (8), Psychology (4), Economics (4), Sport (2), Social Media (1), Computer Vision (1), Library (1), Information Systems (1), Linguistics (1), Computer Networks (1)
user	44	Social Networks (12), Computer Vision (8), Psychology (4), Information Systems (4), Finance & Marketing (4), Social Media (3), Health (2), Sport (2), Music (2), Cardiology (1), Economics (1), Computer Networks (1)
administrator	39	Psychology (7), Social Networks (5), Social Media (4), Cardiology (4), Library (4), Economics (3), Health (2), Endocrinology (2), Radiology (2), Law (2), Computer Networks (2), Transportation (1), Medicine (1)
policymaker	39	Economics (8), Psychology (6), Information Systems (4), Social Work (3), Pharmacology (3), Finance & Marketing (3), Law (2), Health (2), Political Science (1), Computer Vision (1), Transportation (1), Radiology (1), Library (1), Demography (1), Computer Networks (1), Sport (1)
advocate	29	Transportation (4), Finance & Marketing (4), Economics (4), Psychology (3), Pediatrics (2), Radiology (2), Law (2), Linguistics (2), Political Science (1), Health (1), Endocrinology (1), Social Work (1), Pharmacology (1), Information Systems (1)

officer	23	Law (5), Health (3), Social Media (2), Transportation (2), Finance & Marketing (2), Social Networks (2), Psychology (1), Political Science (1), Endocrinology (1), Computer Vision (1), Social Work (1), Economics (1), Computer Networks (1)
healthcare provider	21	Psychology (5), Cardiology (4), Medicine (4), Health (3), Radiology (2), Endocrinology (1), Pharmacology (1), Pediatrics (1)
customer	20	Information Systems (14), Finance & Marketing (5), Economics (1)
patient	19	Health (7), Radiology (4), Cardiology (3), Endocrinology (3), Medicine (2)
product manager	18	Information Systems (3), Social Networks (3), Finance & Marketing (3), Linguistics (3), Music (2), Pediatrics (1), Radiology (1), News (1), Computer Networks (1)
machine learning engineer	16	Computer Vision (3), Information Systems (2), Social Networks (2), Social Work (1), Pharmacology (1), Economics (1), Pediatrics (1), Radiology (1), Psychology (1), Demography (1), Linguistics (1), Computer Networks (1)
public health official	14	Psychology (4), Social Work (4), Health (3), Radiology (2), Pediatrics (1)
quality assurance specialist	13	Computer Vision (3), Health (1), Cardiology (1), Pediatrics (1), Medicine (1), Radiology (1), Economics (1), Information Systems (1), Social Networks (1), Computer Networks (1), Transportation (1)
global citizen	12	Information Systems (4), Economics (2), News (1), Transportation (1), Sport (1), Psychology (1), Music (1), Health (1)
IT specialist	11	Health (4), Finance & Marketing (2), Cardiology (1), Computer Vision (1), Medicine (1), Law (1), Social Networks (1)
strategist	11	Political Science (4), Information Systems (2), Transportation (1), Social Work (1), Social Networks (1), Sport (1), Music (1)
city planner	10	Transportation (8), Social Work (1), Health (1)
HR specialist	10	Psychology (7), Economics (3)
parent	10	Psychology (6), Pediatrics (2), Finance & Marketing (2)
healthcare professional	9	Psychology (3), Health (2), Pediatrics (2), Radiology (2)
journalist	9	News (5), Social Media (1), Information Systems (1), Linguistics (1), Health (1)
marketing professional	9	Information Systems (3), Finance & Marketing (2), Social Media (1), Economics (1), Medicine (1), Music (1)
cybersecurity expert	8	Political Science (3), Transportation (1), Social Media (1), Health (1), Computer Networks (1), Social Networks (1)
librarian	8	Library (3), Demography (2), News (2), Information Systems (1)
person	8	Endocrinology (6), Social Media (1), Psychology (1)
software engineer	8	Transportation (2), Linguistics (2), Endocrinology (1), Information Systems (1), Social Networks (1), Music (1)
citizen	7	Health (2), Political Science (1), Economics (1), Law (1), News (1), Transportation (1)
community leader	7	Social Work (5), Health (2)
consumer	7	Information Systems (2), News (2), Transportation (1), Economics (1), Finance & Marketing (1)
engineer	7	Transportation (5), Health (1), Computer Networks (1)

instructor	7	Finance & Marketing (2), Transportation (1), Cardiology (1), Linguistics (1), Sport (1), Computer Vision (1)
regulator	7	Transportation (3), Finance & Marketing (2), Endocrinology (1), Pharmacology (1)
social worker	7	Economics (3), Psychology (3), Law (1)
UX designer	7	Information Systems (3), Transportation (1), Economics (1), Library (1), Social Networks (1)
caregiver	5	Health (3), Endocrinology (2)
company	5	Pharmacology (2), Transportation (1), Finance & Marketing (1), Psychology (1)
mental health professional	5	Psychology (4), Social Media (1)
security professional	5	Computer Networks (3), Finance & Marketing (2)
CEO	4	Finance & Marketing (3), Pharmacology (1)
decision-maker	4	Transportation (2), Law (1), Computer Networks (1)
financial advisor	4	Economics (2), Finance & Marketing (2)
specialist	4	Computer Vision (2), Cardiology (1), Law (1)
system architect	4	Political Science (1), Computer Vision (1), Library (1), Information Systems (1)
teacher	4	Psychology (3), News (1)
biomedical engineer	3	Medicine (2), Pediatrics (1)
clinical trial coordinator	3	Medicine (2), Pharmacology (1)
data engineer	3	Economics (1), Finance & Marketing (1), Computer Vision (1)
economist	3	Economics (2), Transportation (1)
music streaming service provider	3	Information Systems (2), Music (1)
non-profit organization	3	Economics (2), Psychology (1)
nurse	3	Health (2), Pediatrics (1)
psychologist	3	Psychology (2), Social Networks (1)
scholar	3	Demography (2), Library (1)
student	3	Psychology (1), Library (1), Demography (1)
business owner	2	Transportation (1), Psychology (1)
content moderator	2	Social Media (1), News (1)
executive	2	Transportation (1), Finance & Marketing (1)
fact-checker	2	Linguistics (1), News (1)
government official	2	Economics (1), Law (1)

legal professional	2	Law (1), Psychology (1)
logistics coordinator	2	Political Science (1), Finance & Marketing (1)
manufacturer	2	Transportation (1), Computer Vision (1)
professional	2	Transportation (1), Psychology (1)
system designer	2	Psychology (1), Social Media (1)
technology developer	2	Political Science (1), Health (1)

6. O dataset de equivalência gerado

Arquivo: `**documents/datasets/equivalencia-ustai-synthetic.csv**` — 1260 linhas (1 por user story).

Coluna	Descrição
id	Identificador da user story no UStAI (ex.: A1US1Ge)
abstract	Abstract de origem (A1–A42)
abstract_theme	Tema/sistema de IA descrito pelo abstract
us_num	Número da user story dentro do abstract (US1–US10)
llm	LLM que gerou a user story (Gemini_1.5_flash, Llama 3.1 70b, O1_mini)
role_shorten	Papel declarado — eixo de "domínio" do UStAI (âncora)
user_story	Texto completo da user story
equivalent_domain	Domínio de aplicação equivalente (1 dos 34 do Synthetic)
equivalent_ml_task	ML task âncora <i>fine-grained</i> (1 das 367 do Synthetic) — a técnica que nomeia o estudo
equivalent_ml_task_labels	Conjunto multi-label de labels de alto nível (subconjunto dos 25), separados por <code>;</code> — alvo do classificador de tasks do ReFAIR
num_ml_task_labels	Quantidade de labels multi-label da user story (1 a 25)
domain_confidence	Confiança do mapeamento de domínio (High/Medium/Low)
mapping_notes	Observações para mapeamentos de menor confiança

A partir desse CSV é possível gerar o **PDF** e o **Excel** solicitados (mesmo conteúdo, formatos de apresentação).

7. Exemplos de User Stories por domínio equivalente

Para cada um dos **25 domínios equivalentes** identificados, a tabela abaixo traz **uma user story de exemplo extraída do dataset UStAI** (`UStAI-annotated_v2.pdf`). O exemplo mostra como uma história real do UStAI se conecta ao domínio de aplicação do Synthetic User Stories. Sempre que possível, escolheu-se uma user story cujo `role_shorten` é **específico daquele domínio** (papel de domínio único — ver Seção 5.1), tornando a correspondência mais evidente.

Esses exemplos foram exportados para os arquivos `**documents/datasets/exemplos-us-por-dominio.csv` e `.xlsx**` (1 linha por domínio · 25 linhas), com as colunas: `equivalent_domain`, `num_user_stories_no_dominio`, `equivalent_ml_task`, `equivalent_ml_task_labels`, `num_ml_task_labels`, `domain_confidence`, `id`, `abstract`, `abstract_theme`, `llm`, `role_shorten`, `user_story`.

Dominio equivalente	ID (UStAI)	role_shorten	LLM	Exemplo de User Story
Information Systems	A23US9O1	competitor in the search engine market	O1_mini	As a competitor in the search engine market, I want to analyze how our ranking algorithms compare to others using PAWS, so that I can identify areas for improvement and differentiation.
Psychology	A2US3Ge	therapist	Gemini_1.5_flash	As a therapist working with aggressive children, I want objective data on their physical activity patterns to track progress in anger management and de-escalation techniques.
Transportation	A1US1Ge	driver	Gemini_1.5_flash	As a driver, I want a car that uses AI and real-time data to react faster than humans in emergency situations, reducing my risk of accidents.
Economics	A15US7O1	non-profit organization leader	O1_mini	As a nonprofit organization leader, I want to analyze income prediction data, so that I can develop programs that effectively target and alleviate economic disparities within communities.
Finance & Marketing	A19US1LI	banker	Llama 3.1 70b	As a banker, I want to minimize credit risk and optimize lending decisions, so I need a reliable method to predict credit card default rates using data analytics.
Health	A5US2Ge	lab technician	Gemini_1.5_flash	As a lab technician, I want to easily track the location and status of blood and urine samples using the robot to ensure timely processing.
Computer Vision	A11US2Ge	printing press operator	Gemini_1.5_flash	As a printing press operator, I want the platform to analyze digital templates containing CDPs and predict how they will appear on the final printed product, ensuring accurate replication.
Social Networks	A28US1Ge	single person	Gemini_1.5_flash	As a single person looking for love, I want a dating recommender system that goes beyond self-reported profiles and considers real-life interactions and compatibility.
Cardiology	A9US1O1	cardiologist	O1_mini	As a cardiologist, I want the system to quickly analyze patient data using incremental Fuzzy SVM, so I can receive timely and accurate heart disease diagnoses.
Computer Networks	A34US1Ge	analyst	Gemini_1.5_flash	As a cybersecurity analyst, I want to utilize a DNN-based IDS that can detect and classify new and unpredictable cyberattacks automatically, ensuring our network remains secure against evolving threats.
Demography	A29US1Ge	economic historian	Gemini_1.5_flash	As an economic historian, I want to analyze the long-term effects of early life conditions on individuals' later life outcomes, but I need a reliable way to track individuals across historical census records.
Endocrinology	A10US6Ge	health insurance provider	Gemini_1.5_flash	As a health insurance provider, I want to see if this model can be used to reduce the risk of complications and hospitalizations for people with diabetes.

Law	A30US3Ge	judge or magistrate	Gemini_1.5_flash	As a judge or magistrate, I want to have access to objective and reliable information about a defendant's risk of re-offending to inform sentencing decisions.
Library	A24US6Ge	research funder	Gemini_1.5_flash	As a research funder, I want to identify promising research areas and researchers for potential funding opportunities.
Linguistics	A32US5Ge	social media moderator	Gemini_1.5_flash	As a social media moderator, I want to prioritize content for review based on the intensity of a user's stance (e.g., strongly against vs. mildly disagreeing).
Medicine	A17US1Ge	sleep specialist	Gemini_1.5_flash	As a sleep specialist, I want to use an advanced prediction system to forecast episodes of apnea and hypopnea, enabling me to tailor treatment plans more effectively for individual patients.
Music	A39US1Ge	music enthusiast	Gemini_1.5_flash	As a music enthusiast, I want a tool that can predict the success of my playlists based on musical characteristics, allowing me to refine my curation and increase engagement.
News	A33US2LI	news agency	Llama 3.1 70b	As a news agency, I want to be able to search for precise facts described in news articles, so that I can verify information and ensure the accuracy of our reporting.
Pediatrics	A16US1Ge	pediatrician	Gemini_1.5_flash	As a pediatrician, I want a non-invasive tool that can analyze a newborn's cry and help me identify potential health issues early on.
Pharmacology	A13US2Ge	pharmaceutical company scientist	Gemini_1.5_flash	As a pharmaceutical company scientist, I want the platform to predict potential combinations of existing drugs that may have a synergistic effect in treating AD.
Political Science	A4US1Ge	military commander	Gemini_1.5_flash	As a military leader, I want to leverage AI for improved situational awareness and faster decision-making on the battlefield to gain a tactical advantage.
Radiology	A18US1Ge	radiologist	Gemini_1.5_flash	As a radiologist, I want a tool that can assist me in diagnosing TB disease from chest X-rays, improving accuracy and efficiency in my workflow.
Social Media	A3US2Ge	moderator	Gemini_1.5_flash	As a content moderator, I want a tool that identifies potentially aggressive tweets with high accuracy so that I can prioritize them for review and take appropriate action.
Social Work	A12US7O1	grant writer	O1_mini	As a grant writer, I want to utilize the FVVI's standardized risk assessments, so that I can secure funding for targeted violence prevention programs.
Sport	A37US1Ge	sports fan	Gemini_1.5_flash	As a casual sports fan, I want a platform that uses data and analytics to help me make informed decisions about which players to select for my fantasy team.

Como ler a tabela. Cada linha é uma user story *real* do UStAI; a coluna **Domínio equivalente** indica o domínio do Synthetic User Stories ao qual aquela história (e todo o seu *abstract*) pertence, conforme a

equivalência da Seção 4. O `id` permite localizar a história no dataset completo (`equivalencia-ustai-synthetic.csv`).

8. Limitações

- **Domínio fixado por abstract.* Como cada abstract descreve um só sistema, o domínio é constante entre suas 30 user stories. Eventuais user stories "fora do tema" geradas pelos LLMs (ex.: um cybersecurity expert falando de rede em um abstract de saúde) mantêm o domínio do abstract*.
- **6 abstracts de baixa confiança** (A4, A12, A21, A22, A31, A38): não há domínio perfeito entre os 34 — ver notas na Seção 4.
- **ML task multi-label por abstract.* Tanto a task âncora quanto o conjunto de labels multi-label são fixados por abstract* (a técnica central do estudo de origem); user stories individuais podem mencionar tarefas auxiliares. A expansão para os 25 labels usa a aba `task augmentation mapping` do ReFAIR — logo, herda as escolhas dessa tabela.
- **Abstracts com técnica de uso geral.** Quando a task âncora é uma técnica genérica, o conjunto multi-label fica pouco discriminativo: `deep learning` (A13) expande para os **25** labels; `multi-task learning` (A32) e `deep neural network` (A34) para **17**; `artificial neural network` (A17) para **16**. Nesses casos o conjunto reflete a versatilidade da técnica, não uma restrição forte de task.
- Vocabulário-alvo: 34 domínios, 367 ML tasks *fine-grained* e 25 labels de ML task multi-label, extraídos de `Synthetic User Stories.xlsx` (abas *Dataset*, *Domains*, *ML Dictionary*, *task augmentation mapping*).